

TRND model logic

How a pick page is made, block by block. Reference as of 2026-09-13. Every rule here is what the code does today; file names are given so a rule can be checked.

1. The page, top to bottom

A pick page is assembled from stored rows. Nothing on it is computed when the page loads, and no model is called on the request path. The weekly job writes every row; the page reads them.

Block	What it is	Made by
Title	The term the pick is about ("why is my hair so dry").	The ranking picked the term; stored on the pick.
Signal read	Four scores 0-100 with a confidence each, and the components behind them.	The four scorers, lib/scoring/*, at ranking time; stored as signal_scores.
Opportunity Grade	Letter and 0-100 score, stars, and which signals were left out.	combineSignals in lib/scoring/model.ts.
Demand card	One metric: label, value, delta with its window, 30-point sparkline.	pickMetric in lib/picks/metric.ts. Never a model.
The finding	One sentence: what customers say against what the product page says.	Gemini, held to a schema and validated. lib/ai/pick-writer.ts.
Guardrail	What the ad must never say (platform policy, brand risk).	Gemini, same call as the finding.
The bet	What to run, budget, duration, kill rule.	Numbers by lib/picks/bet.ts; the "what to run" sentence by Gemini.
Scripts	Three variants: thesis, hook, beats, CTA, length.	Gemini; length from the winning short-form length on the term.
Why this pick	Evidence claims per signal with links to the source page.	lib/picks/evidence.ts from stored facts. Never a model.

2. Where the data comes from

Every source is an adapter in lib/signals/adapters. Each writes rows to the signals table (one per source, term and day) and, where the source has daily shape, points to the series table. The daily cron runs them for every brand's watch terms; a fresh signup runs them for its own terms during the first-week chain.

Source	Reads	Metric type	Notes
DataForSEO	Google monthly search volume per term	search_volume	The anchor for size: the demand card's 2,400 is this. Monthly totals, so its delta is month over month.
Google Trends	Daily interest index 0-100, up to a year	search_interest + series	The sparkline and the lifecycle read. An index, so never shown as a count.
Google autocomplete	Suggestions and buying modifiers	search_intent	Customer words for the intent read.
TikTok per term (Apify)	Recent videos on the term: views, shares, saves, captions	shortform_views + series	Paid. Corpus keeps the month's posts. Shares and saves per view feed intent.

Source	Reads	Metric type	Notes
TikTok Creative Center	National board per industry	conversation	Free, three rows per industry.
YouTube Shorts	Shorts per term	shortform_views	Key-gated, unit budget.
X	Posts per term, last 7 days	posts	Key-gated.
Reddit	Top posts in the customer's subreddits	conversation	Titles become customer words.
Google News	Stories per term	news_coverage	Context only.
Own accounts (Apify)	Instagram, TikTok, Facebook: 30 recent posts each	social_posts table	Handles read from the brand's own site footer. Refreshed every 48h.
Rival accounts (Apify)	Same read for each watched competitor	social_posts table	A handle another workspace read this week is copied, not re-scraped.
Meta Ad Library (Apify)	Each rival's active ads, running days, copy	competitor_reads (ads)	A failed read is stored as nothing, never as "no ads".
Google Ads Transparency	Each rival's Google ads by domain	competitor_reads (google_ads)	Free. Counts as ad evidence.
Brand and rival websites	Products, prices, social links, copy	services, competitors	Sites behind a bot wall are read through a rendering reader proxy.
Google Places	Reviews and ratings	reviews	Local businesses only.

3. The weekly ranking

lib/recommend/recommend.ts. Once a week (and after onboarding, and on the owner's re-rank) each brand's category pool, every signal row from the last 14 days, is turned into at most five opportunities.

- **Pool.** Every signal in the brand's category from the last 14 days, plus the brand's own watch terms.
- **Relevance judge.** Gemini reads the brand's analysis (the brief) and judges each term's fit to what the brand sells, 0-1. Without a key, a concept map per category gives the same shape of verdict. Terms judged irrelevant are dropped unless the whole pool is.
- **Fit gate.** A term with judged fit under 0.4 is a Hold whatever its other signals: an espresso-martini trend at a smokehouse is never a pick.
- **Grade.** Every surviving term is scored by the four signals and combined into the Opportunity Grade (section 5).
- **Top five.** Ordered by grade score. A Hold is never stored and never becomes a pick. Rows that fell out of this ranking are deleted unless a campaign holds them, so yesterday's row cannot keep a seat.
- **Baselines.** Each graded term writes today's raw readings to signal_readings. After eight prior days exist, percentiles replace the absolute curves.

4. The four signals

Contract in lib/scoring/model.ts. Each signal returns a 0-100 score, a confidence (high, medium, low), and three components with a plain-English detail. Two rules: score relative when a baseline exists (percentile against the brand's trailing 90 days, category for Culture), and never fake a number: a low-confidence signal is left out of the grade with a visible note, and its weight goes to the others. A component with no data is null and its weight spreads across the ones that have data.

Customer (lib/scoring/customer.ts)

Component	Weight	How it is read
Volume	40	This week's level for the term, ranked against the brand's trailing readings. Until eight readings exist, an absolute curve per kind: searches 10 is 25, 100 is 50, 1,000 is 75, 10,000 is 100; short-form views one step higher; a Trends index is used as is. Discounted by persona match (how closely the term is how the brand's target customer talks, from the brief, blended with own-post engagement).
Intent	35	Customer texts on the term (autocomplete phrasings, Reddit titles, X posts, short-form captions from the month) classified by a fixed phrase list into purchase intent, pain point, complaint, curiosity, unrelated. Needs three or more texts. Blended 60/40 with shares and saves per view on the term's short-form when that read has 2,000 or more views: 0.1% is 25, 0.5% is 50, 1% is 75, 2% is 100.
Velocity	25	Last 7 days of the term's daily series against the last 30 (needs 14 points). Half the month's pace is 0, level is 50, double is 100.

Confidence: low only with no persona and no level, or nothing read at all. High needs a persona, a ranked baseline, five or more customer texts and two weeks of daily points.

Culture (lib/scoring/culture.ts, lifecycle.ts)

Component	Weight	How it is read
Category growth	40	Median week-over-week move across the category's short-form, Trends and TikTok board reads, ranked against the category's trailing growth readings. Until then: -50% is 0, flat is 40, +50% is 100. One number per category, so it does not separate terms within a brand.
Seasonal fit	30	With a year of the term's own series: how the same five weeks last year ran against that year's monthly norm (0.6x is 10, the norm is 50, 1.5x is 95). Otherwise the category calendar: in window 90, within three weeks 70, out of season 30.
Lifecycle	30	The stage read from the shape of the 90-day series: emerging 90, growing 80, flat 50, peaking 35, declining 15. Reads week over week, against the prior month, and (with 60+ days) the last two weeks against the first month, so a plateau reached three weeks ago still reads as a rise being ridden.

Competitive (lib/scoring/competitive.ts)

Component	Weight	How it is read
Whitespace	45	Share of watched rivals NOT running this angle now (posts or ads on the term in the last 14 days). None of three is 100; all three is 0.
Saturation trend	30	Rivals on the angle now against 15-45 days ago: fewer is 80, same is 50, more is 20.
Competitor weakness	25	Share of their ads on the angle that look weak: pulled inside a week, or a post under half the rival's own median engagement.

Watched rivals: everyone over the directness bar (0.5), or the three nearest at 0.3 or above when nobody clears it. A rival counts as read only when a post or an ad of theirs was actually seen; an empty ad read is not a read. Zero rivals: low, a placeholder, and an "Add competitors" link. High needs three rivals read, a prior window, and twelve or more items behind the call. Directness (lib/intel/direct.ts): for an online brand, the share of the brand's product words (the words on most of its items plus its category) on the rival's site, and the share of the customer lexicon the rival's site speaks; a site that says the whole category phrase sells the product.

Brand (lib/scoring/brand.ts)

Component	Weight	How it is read
Past ad performance	40	Past ads on the term (imported Meta or Google exports, or a synced ad account): their click-through against the account average, ranked against the brand's other ads. Blank without ad history.
Catalog fit	30	The judged fit of the term to what the brand sells (the same judgment the fit gate uses), minus 20 outside the price band, minus 15 on thin margin, 0 out of stock. Stock and margin are null today: no feed.
Organic traction	30	The brand's own posts on the term against its median post. Posts on the term: word-stem match (a one or two word term needs every stem; a longer one may miss one). Engagement is likes plus comments plus shares. Median of the on-term posts over the median of all posts: half is 0, equal is 50, double is 100. Five or more posts read and none on the term is 30, a reading rather than a gap.

Confidence: medium with two components, or with a judged catalog fit alone. High needs ten or more ads, all three components and a ranked lift. Worked example, Jolie, "why is my hair so dry": ad performance blank; catalog fit 100; three of 42 posts mention dry hair, median engagement 349 against 199 across all posts, ratio 1.75, organic 92. Brand = $(30 \times 100 + 30 \times 92) / 60 = 96$.

5. The Opportunity Grade

- **Weights.** Customer 35, Brand 25, Culture 20, Competitive 20, used when every signal is medium or better.
- **Redistribution.** A low-confidence signal is excluded and the remaining weights are renormalized to 100. The page says which were left out and why. With nothing above low, the term is a Hold.
- **Bands.** 90-100 A+, 80-89 A, 70-79 B+, 60-69 B, 50-59 C, under 50 Hold. A Hold never becomes a pick.
- **Stars.** The score's fifth, rounded down: 66 is three stars.
- **Stored.** grade, grade_score and the four signal blobs are written on the opportunity and copied onto the pick, so the page shows what the brand was graded on that week.

6. The demand card

lib/picks/metric.ts. A pick shows exactly one number, chosen by code and never restated elsewhere on the page.

- **Which signal.** The pick's own signal, unless it is a short-form read under 5,000 views, which hands over to a search read on the same term. With no measured change at all there is no metric and the pick stays a draft.
- **Label.** By source: "Searches for" (DataForSEO, Trends, autocomplete), "TikTok views on", "Posts on X about", and so on. A widened read names the term actually measured.
- **Value.** Only when the value is a count someone could check: monthly searches, views, posts. A Trends index is never shown as a value.
- **Delta and window.** This week's measured change when there is one, else the 30-day trajectory. DataForSEO arrives as monthly totals, so its delta is labelled "over 30 days" even when it is the week's read.
- **Sparkline.** The last 30 daily points of the term's series, oldest first, unsmoothed. When a term has both a Trends index and raw volumes, the index is kept because it has daily shape.

The card in the question: "why is my hair so dry" is a DataForSEO term. Value 2,400 is its monthly Google search volume. "0% over 30 days" is that volume month over month. The line is the Google Trends daily index for the term over the last 30 days, which moves while the monthly total does not; that is why a flat delta can sit

beside a jagged line.

7. The finding, the bet, the scripts

- **Numbers first.** The bet is computed before any model runs (`lib/picks/bet.ts`). Online brand: budget is 5% of the monthly spend band's floor spread over seven days, run as a five-day test. Local: the price band's daily floor over six days. Kill rule, best evidence first: cost per purchase 30% over the account's own average by day 3; else cost per click; else the category's click-through benchmark.
- **Facts second.** Evidence rows are built from stored facts (`lib/picks/evidence.ts`): the customer's words, the winning short-form and its length, rival moves and ads, the brand's own best posts. Every claim links to a source page. A signal with nothing behind it writes no rows.
- **Words last.** One Gemini Pro call (`lib/ai/pick-writer.ts`, prompt in `lib/ai/prompts/pick.ts`) receives the brand, the term, the direction of the metric without its figure, the matched product and the whole catalog, the brief, the signal brief and the evidence claims. It returns the finding, what to run, the guardrail and three scripts, held to a response schema.
- **Validation.** The finding must name the term and must not repeat the metric figure; the item in the finding must be the item the bet runs; if the brand already uses the customer's words the finding must say what the page leads with instead. A violation retries once on Pro, then on Flash; a pick that still fails is stored as a draft the list never shows.
- **Script length.** The median length of the winning short-form on the term, clamped to 15-45 seconds.

8. When it all runs

- **First week.** `/api/jobs/week` runs one stage per invocation and hands off to itself: analysis, market scan, own and rival reads, ranking, picks. Each read stage has a 200-second budget and resumes where it stopped. The picks page reads what the week still needs from the database and asks for the next stage; it never runs the chain.
- **Daily.** Market ingest at 09:00 UTC, intel (own accounts, rivals, ads) at 09:30 with its own budget, results sync at 11:00.
- **Weekly.** Ranking Monday 10:00, picks 10:20, the email 10:30.
- **Learning loop, v1.** A completed run's impressions and clicks are written into the brand's own ad history, so the next Brand read is provable rather than estimated. Learned per-brand weights are v2, after 30-50 completed runs.

9. Honesty rules that hold everywhere

- A number on the page is computed by code from stored rows; a model only writes words.
- A signal with thin data says so in its detail line rather than defaulting to a credible-looking middle.
- A read that failed upstream is stored as nothing, never as zero.
- The metric figure appears once. Claims and copy that would repeat it are rejected.
- Every claim under "Why this pick" links to the page it was read from.